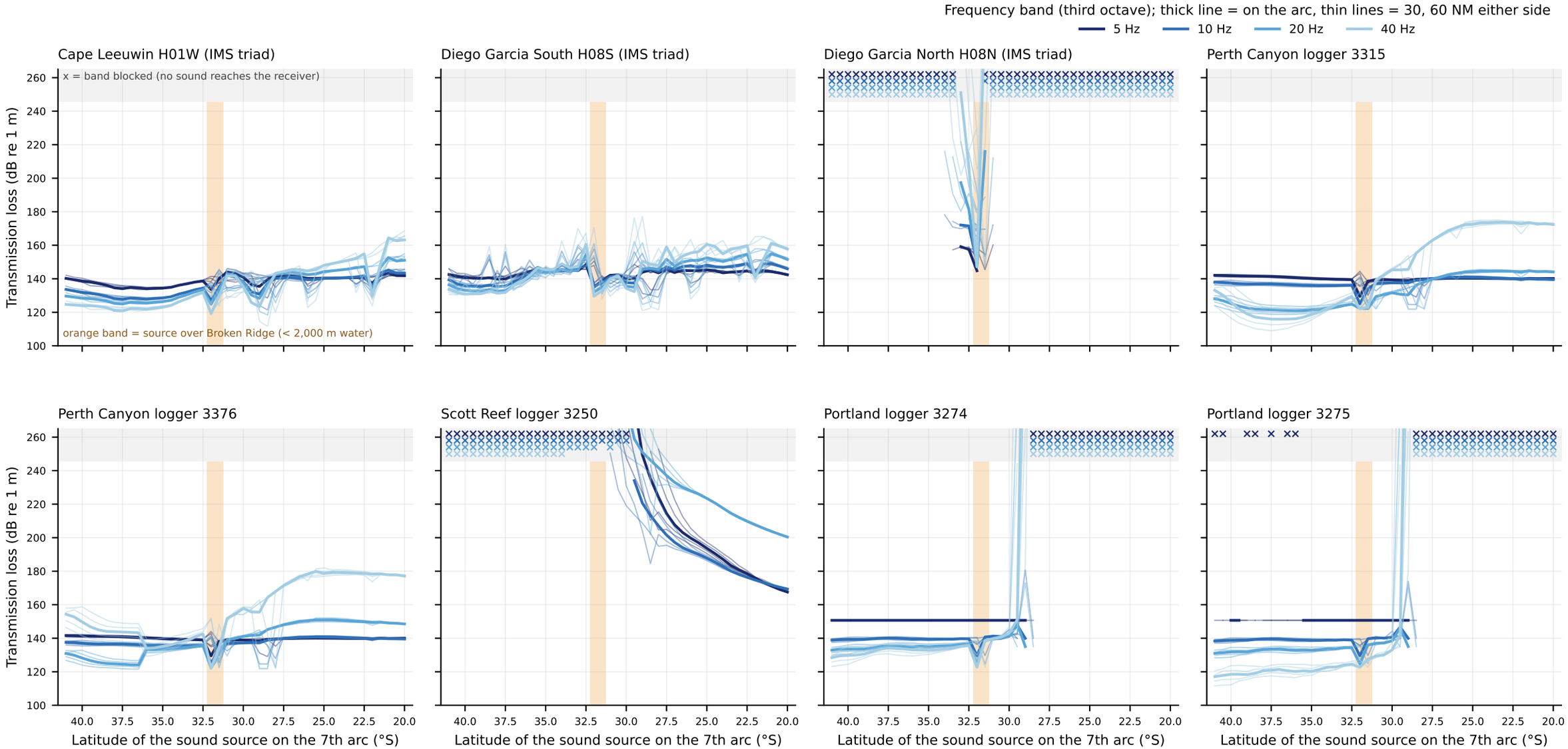
Sound transmission loss from the 7th arc to each receiver, by source latitude: Diego Garcia North is blocked from almost everywhere, Scott Reef from south of 29.5°S



Plain-language note: Each line shows how much the sound from an impact on the 7th arc is reduced on its way to one receiver, for sources from 41°S to 20°S. A higher value means a weaker arriving sound. A cross means the sea floor is too shallow somewhere on the path for that frequency, so the model sends no sound through. The thin lines are for sources 30 (and, where drawn, 60) nautical miles west and east of the arc. In the orange band the source sits over Broken Ridge in water less than 2,000 m deep; the model then lets more sound into the deep channel, and this part of the model is not validated. Values are estimates from a model of the ocean; they are not measurements, and the model is not yet calibrated.

Technical note: KRAKEN adiabatic incoherent modes along the WGS84 geodesic, GEBCO_2026 bathymetry (5 km profiles, median of 3x3 cells), WOA23 2005-2014 March sound speed, hard bottom (cp 1650, rho 1.9, 0.5 dB/lambda), Francois-Garrison absorption, spherical correction; source 10 m; receiver at the triad hydrophone depth (IMS) or 2 m above the seabed (IMOS). IMS values have the module's RAM delta subtracted (ims_ram_delta.csv, H01W/H08S only); IMOS values carry the module's RAM correction (ram_tl_check rule 2; none for 3250). Blocked = KRAKEN finds no propagating mode on a profile (or land on track, or zero field at the receiver); at Portland the 5 Hz line is the RAM reference value that rule 2 assigns to a no-mode band. ridge_source rows (orange) are flagged (reach gap G-N2/K5); at Portland their RAM-corrected value is floored at the reference RAM value (tl_grid_table.py). Arc = zero contour of the project's arc-distance field (searched-areas arc-grid.npz); offsets in that field. Caveats C1 (slope misfit +10.4/+7.4 dB per octave at H01W/H08S) and C2 (coupling not calibrated). Grid: passes A+B (offsets -30/0/+30 NM, mandated) and C (±60 NM, supplementary); module commits fc825c0, 4392091.